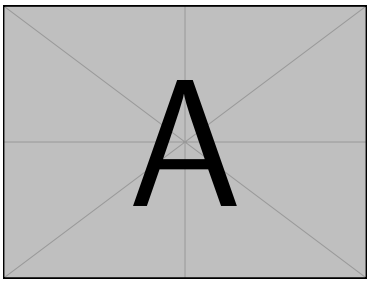



# LaTeX Complete Reference

A Comprehensive Learning Guide

New Learner

April 1, 2026

# Contents


**1 Document Structure & Document Classes** **5**
1.1 Document Classes . . . 5
1.1.1 Document Class Options . . . 5
1.2 Sectioning Commands . . . 6
1.2.1 Starred Versions . . . 7
1.3 Input and Include . . . 7
1.4 Comments . . . 7
1.5 Special Characters . . . 7

**2 Text Formatting & Fonts** **9**
2.1 Font Styles . . . 9
2.1.1 Examples . . . 9
2.2 Font Sizes . . . 10
2.3 Text Alignment . . . 10
2.3.1 Centering . . . 10
2.3.2 Flush Left (Ragged Right) . . . 10
2.3.3 Flush Right (Ragged Left) . . . 11
2.4 Line & Paragraph Spacing . . . 11
2.4.1 Vertical Spacing . . . 11
2.4.2 Horizontal Spacing . . . 12
2.5 Text Decoration & Color . . . 12
2.5.1 Underline and Overline . . . 12
2.5.2 Strikethrough (requires ulem package) . . . 12
2.5.3 Text Color (xcolor package) . . . 12
2.6 Accents and Special Characters . . . 13
2.7 Dashes and Quotes . . . 13

**3 Lists** **15**
3.1 Itemize (Bulleted Lists) . . . 15
3.1.1 Nested Itemize . . . 15
3.2 Enumerate (Numbered Lists) . . . 16
3.2.1 Nested Enumerate . . . 16
3.3 Description Lists . . . 16
3.4 Customizing Lists with enumitem . . . 17
3.4.1 Custom Labels . . . 17
3.4.2 Custom Bullet Symbols . . . 17
3.4.3 Spacing Control . . . 18
3.5 Inline Lists . . . 19
3.6 List Summary . . . 19

**4 Tables** **20**
4.1 Basic Table (tabular) . . . 20
4.1.1 Column Types . . . 20
4.1.2 Basic Example . . . 20
4.2 Table Syntax . . . 20
4.3 Professional Tables with booktabs . . . 21
4.4 Multi-column and Multi-row . . . 22
4.4.1 Multi-column (`\multicolumn`) . . . 22
4.4.2 Multi-row (multirow package) . . . 22
4.5 Fixed-width Tables (tabularx) . . . 22
4.6 Multi-page Tables (longtable) . . . 23
4.7 Table Placement (Float) . . . 24
4.8 Colored Tables . . . 24
4.9 Table Quick Reference . . . 25

**5 Mathematics** **26**
5.1 Math Modes . . . 26
5.2 Superscripts and Subscripts . . . 27
5.3 Fractions . . . 27
5.4 Roots . . . 27
5.5 Sums, Products, Integrals . . . 27
5.6 Greek Letters . . . 28
5.7 Mathematical Operators and Functions . . . 28
5.8 Relations and Binary Operators . . . 29
5.9 Arrows . . . 29
5.10 Matrices . . . 29
5.11 Multi-line Equations (align) . . . 30
5.12 Cases and Piecewise Functions . . . 31
5.13 Accents in Math Mode . . . 32
5.14 Spacing in Math Mode . . . 32
5.15 Text Inside Math Mode . . . 32
5.16 Theorems and Proofs . . . 33
5.17 Comprehensive Math Example . . . 33

**6 Figures & Images** **34**
6.1 Including Images (graphicx) . . . 34
6.2 Figure Environment (Float) . . . 34
6.2.1 Referencing Figures . . . 35
6.3 Subfigures (subcaption package) . . . 35
6.4 Image Size Options . . . 36
6.5 Wrapping Text Around Images . . . 36
6.6 Drawing Simple Graphics with TikZ (Preview) . . . 36

**7 Cross References & Hyperlinks** **38**
7.1 Labels and References . . . 38
7.1.1 Label Naming Conventions . . . 39
7.1.2 Pageref and Nameref . . . 39
7.2 Hyperlinks (hyperref package) . . . 39
7.3 Table of Contents . . . 40

7.3.1 Custom TOC Entry . . . 40

**8 Bibliography & Citations** **41**
8.1 Basic Bibliography with thebibliography . . . 41
8.1.1 Citation Commands . . . 42
8.2 BibTeX with .bib Files . . . 42
8.2.1 The .bib File . . . 42
8.2.2 Entry Types . . . 43
8.2.3 Common Fields . . . 44
8.3 Using BibTeX . . . 44
8.3.1 Bibliography Styles . . . 44
8.4 Biblatex (Modern Approach) . . . 45
8.5 Example Bibliography . . . 45

**9 Code Listings & Verbatim Text** **47**
9.1 Verbatim Text . . . 47
9.1.1 Inline Verbatim Examples . . . 47
9.2 Code Listings (listings package) . . . 48
9.2.1 Basic Listing . . . 48
9.2.2 Listing with Language . . . 48
9.2.3 Java Example . . . 48
9.2.4 C/C++ Example . . . 48
9.2.5 JavaScript Example . . . 49
9.3 Listing Styles and Configuration . . . 49
9.4 Inline Code . . . 50
9.5 Supported Languages . . . 50

**10 TikZ Graphics** **51**
10.1 Basic Shapes . . . 51
10.2 Lines and Paths . . . 51
10.3 Filling and Colors . . . 52
10.4 Nodes and Labels . . . 52
10.5 Flowchart . . . 53
10.6 Plots with pgfplots . . . 53
10.7 Bar Chart . . . 54
10.8 TikZ Styles . . . 54

**11 Page Layout & Headers/Footers** **56**
11.1 Page Geometry (geometry package) . . . 56
11.1.1 Understanding Page Layout . . . 57
11.2 Headers and Footers (fancyhdr) . . . 57
11.2.1 Position Codes . . . 58
11.3 Page Numbering . . . 58
11.4 Multiple Columns . . . 58
11.5 Page Breaks . . . 59
11.6 Paragraph Control . . . 59

**12 Floating Bodies & Captions** **60**
12.1 What Are Floats? . . . 60

12.2 Placement Specifiers . . . 60
12.3 Captions . . . 61
12.3.1 Caption Styles (caption package) . . . 61
12.4 Side Captions . . . 62
12.5 Float Barriers . . . 62
12.6 Controlling Float Counters . . . 62

**13 Essential Packages & Their Uses** **63**
13.1 Package Management . . . 63
13.2 Essential Package Reference . . . 63
13.3 Package Conflicts and Loading Order . . . 64
13.4 SI Units (siunitx) . . . 65
13.5 Creating Your Own Package . . . 66

**14 Custom Commands & Environments** **67**
14.1 New Commands (`\newcommand`) . . . 67
14.1.1 Practical Examples . . . 67
14.2 Renaming Commands (`\renewcommand`) . . . 68
14.3 New Environments . . . 68
14.4 Lengths and Counters . . . 68
14.4.1 Defining Lengths . . . 68
14.4.2 Standard Lengths . . . 69
14.4.3 Counters . . . 69
14.5 Conditional Commands . . . 69
14.6 Boxes . . . 70

**15 Advanced Topics** **72**
15.1 Tables of Contents for Non-Standard Sections . . . 72
15.2 Index . . . 72
15.3 Glossary / Acronyms . . . 73
15.4 Watermarks . . . 73
15.5 Conditional Compilation . . . 74
15.6 External Files and Shell Escape . . . 74
15.7 Compilation Tools . . . 74
15.7.1 latexmk . . . 74
15.7.2 Compilation Engines . . . 75
15.7.3 Using System Fonts (XeLaTeX/LuaLaTeX) . . . 75
15.8 Beamer Presentations (Overview) . . . 75
15.9 Multilingual Support . . . 76
15.10 Error Handling . . . 77

**A Quick Reference Card** **78**
A.1 Document Skeleton . . . 78
A.2 Symbol Quick Reference . . . 80
A.3 Useful Commands Cheat Sheet . . . 81
A.4 Compilation Workflow . . . 81

# Chapter 1

# Document Structure & Document Classes

Every LaTeX document consists of two main parts:

1. **Preamble** everything before `\begin{document}`
2. **Body** the content between `\begin{document}` and `\end{document}`

## 1.1 Document Classes

The `\documentclass` command must be the very first command. It defines the type of document.

| Class | Purpose | Sectioning |
|---|---|---|
| `article` | Short documents, journal papers | section, subsection |
| `report` | Longer documents, theses | chapter, section |
| `book` | Full books | part, chapter, section |
| `letter` | Letters | |
| `beamer` | Presentations/slides | frame |
| `scrartcl` | KOMA-Script article | section |
| `scrreprt` | KOMA-Script report | chapter |
| `scrbook` | KOMA-Script book | part, chapter |

Table 1.1: Common document classes in LaTeX

### 1.1.1 Document Class Options

Class options go in square brackets: `\documentclass[options]{class}`

| Option | Effect |
|---|---|
| `12pt` | Sets font size (10pt, 11pt, 12pt) |
| `a4paper` | Sets paper size (a4paper, letterpaper, legalpaper) |
| `twoside` | Formats for two-sided printing |
| `oneside` | Formats for one-sided printing (default) |
| `landscape` | Landscape orientation |
| `draft` | Draft mode marks overfull boxes |
| `final` | Final mode (default) |
| `twocolumn` | Two-column layout |
| `titlepage` | Title on a separate page |
| `openright` | Chapters start on right-hand page |
| `leqno` | Equation numbers on left |
| `fleqn` | Left-aligned equations |

Table 1.2: Common document class options

```latex
\documentclass[12pt, a4paper]{article}

% === PREAMBLE ===
% Load packages, define commands, set options here
\usepackage[utf8]{inputenc}
\usepackage{graphicx}
\title{My First Document}
\author{Author Name}

% === BODY ===
\begin{document}
\maketitle
Hello, World!
\end{document}
```

Listing 1.1: Basic document structure

## 1.2 Sectioning Commands

LaTeX provides a hierarchy of sectioning commands:

```latex
\part{Part Title}           % Level -1 (book/report only)
\chapter{Chapter Title}     % Level 0  (book/report only)
\section{Section Title}     % Level 1
\subsection{Subsection}     % Level 2
\subsubsection{Sub-sub}     % Level 3
\paragraph{Paragraph}       % Level 4
\subparagraph{Subparagraph}% Level 5
```

Listing 1.2: Sectioning hierarchy

### 1.2.1 Starred Versions

Each sectioning command has a starred version (e.g., `\section*{Title}`) that:

- Does *not* add a number
- Does *not* appear in the table of contents

## 1.3 Input and Include

Split large documents into multiple files:

```
% \input{file}      -- inserts file content inline (no page break)
% \include{file}    -- inserts file with \clearpage before and
   after
%                      (cannot be nested)

\input{chapters/chapter1}      % Preferred for small files
\include{chapters/chapter2}    % Preferred for chapters
```

Listing 1.3: File inclusion commands

| Feature | `\input` | `\include` |
|---|---|---|
| Page breaks | No | Yes |
| Can be nested | Yes | No |
| Selective compilation | No | Yes (`\includeonly`) |

Table 1.3: Comparison of `\input` vs `\include`

## 1.4 Comments

```
% This is a single-line comment.
% Everything after % on the line is ignored.

% For multi-line comments, use the verbatim package:
\usepackage{verbatim}
\begin{comment}
This entire block is commented out.
Line two of the comment.
Line three.
\end{comment}
```

Listing 1.4: Comments in LaTeX

## 1.5 Special Characters

The following characters are **reserved** in LaTeX and need special handling:

| **Character** | **Purpose** | **To Print It** |
|---|---|---|
| `#` | Macro parameter | `\#` |
| `$` | Math mode | `\$` |
| `%` | Comment | `\%` |
| `&` | Table column sep. | `\&` |
| `_` | Subscript (math) | `\_` |
| `{` | Group begin | `\{` |
| `}` | Group end | `\}` |
| `~` | Non-breaking space | `\~{}` or `\textasciitilde` |
| `^` | Circumflex accent | `\^{}` or `\textasciicircum` |
| `\` | Command prefix | `\textbackslash` |

Table 1.4: Special characters in LaTeX

*Note: Accented characters can be typed directly in UTF-8: é, ü, ñ, ç, etc.*

# Chapter 2

# Text Formatting & Fonts

## 2.1 Font Styles

LaTeX provides command pairs for font switching one with *declaration syntax* (affects all following text) and one with *command syntax* (takes an argument).

| Style | Declaration | Command |
|---|---|---|
| **Bold** | `\bfseries` | `\textbf{...}` |
| *Italic* | `\itshape` | `\textit{...}` |
| *Slanted* | `\slshape` | `\textsl{...}` |
| Monospace | `\ttfamily` | `\texttt{...}` |
| Sans-serif | `\sffamily` | `\textsf{...}` |
| Small Caps | `\scshape` | `\textsc{...}` |
| Medium | `\mdseries` | `\textmd{...}` |
| Upright | `\upshape` | `\textup{...}` |

Table 2.1: Font style commands

### 2.1.1 Examples

- **Bold text** with `\textbf{Bold text}`
- *Italic text* with `\textit{Italic text}`
- Typewriter text with `\texttt{Typewriter text}`
- Sans-serif text with `\textsf{Sans-serif text}`
- Small Capitals with `\textsc{Small Capitals}`
- *Slanted text* with `\textsl{Slanted text}`
- *Emphasized text* `\emph{}` toggles italic inside roman and roman inside italic
- *This is* nested *emphasis* `\emph{This is \emph{nested} emphasis}`

## 2.2 Font Sizes

Size commands are *declarations* they affect everything until the end of the group:

| Command | Result |
|---|---|
| `\tiny` | Sample text |
| `\scriptsize` | Sample text |
| `\footnotesize` | Sample text |
| `\small` | Sample text |
| `\normalsize` | Sample text |
| `\large` | Sample text |
| `\Large` | Sample text |
| `\LARGE` | Sample text |
| `\huge` | Sample text |
| `\Huge` | Sample text |

Table 2.2: Font size commands (relative to base document font size)

```
{\large This text is large.}
{\Large\bfseries This text is large and bold.}
```

Listing 2.1: Font size usage

## 2.3 Text Alignment

### 2.3.1 Centering

```
% Environment form:
\begin{center}
    This text is centered.
\end{center}

% Declaration form (affects rest of group):
{\centering This text is centered.\par}
```

Listing 2.2: Centering text

This is centered text using the `center` environment.

### 2.3.2 Flush Left (Ragged Right)

```
\begin{flushleft}
    This text is left-aligned.
\end{flushleft}
```

Listing 2.3: Left-aligned text

### 2.3.3 Flush Right (Ragged Left)

```latex
\begin{flushright}
    This text is right-aligned.
\end{flushright}
```

Listing 2.4: Right-aligned text

This is right-aligned text.

## 2.4 Line & Paragraph Spacing

```latex
% Single spacing (default)
% Double spacing:
\usepackage{setspace}
\doublespacing

% 1.5 spacing:
\onehalfspacing

% Custom spacing:
\setstretch{1.25}

% Line break (new line, same paragraph):
Line one.\\
Line two.

% Line break with extra space:
Line one.\\[12pt]
Line two with 12pt gap above.

% New paragraph (blank line in source):
Paragraph one.

Paragraph two.

% Horizontal spacing:
Word\hspace{1cm}Word          % 1cm horizontal space
Word\hspace{\fill}Word        % Stretchy space (pushes to margins
    )
Word\hfill Word               % Shorthand for \hspace{\fill}
```

Listing 2.5: Line spacing

### 2.4.1 Vertical Spacing

```latex
\vspace{1cm}            % 1cm vertical space
\vspace*{1cm}           % Forced space (even at page top)
\vspace{\fill}          % Stretchy vertical space
\vfill                  % Shorthand for \vspace{\fill}
\bigskip                % Large vertical skip (~12pt)
```

```latex
\medskip                % Medium vertical skip (~6pt)
\smallskip              % Small vertical skip (~3pt)
```

Listing 2.6: Vertical spacing

### 2.4.2 Horizontal Spacing

| Command | Width | Description |
|---|---|---|
| `\,` or `\thinspace` | ∼3/18 quad | Thin space |
| `\:` or `\medspace` | ∼4/18 quad | Medium space |
| `\;` or `\thickspace` | ∼5/18 quad | Thick space |
| `\!` | −3/18 quad | Negative thin space |
| `\enspace` | 0.5em | En-space |
| `\quad` | 1em | Quad space |
| `\qquad` | 2em | Double quad |
| `\~{}` | variable | Non-breaking space |
| `\hspace{length}` | variable | Custom horizontal space |

Table 2.3: Horizontal spacing commands

## 2.5 Text Decoration & Color

### 2.5.1 Underline and Overline

```latex
\underline{Underlined text}
\overline{Overlined text}          % Works in math mode too
```

Listing 2.7: Underline and overline

<u>Underlined text example.</u>

### 2.5.2 Strikethrough (requires ulem package)

```latex
\usepackage[normalem]{ulem}

\sout{Strikethrough text}
\uwave{Wavy underline}
\xout{Crossed out with /}
```

Listing 2.8: Strikethrough

### 2.5.3 Text Color (xcolor package)

```latex
\textcolor{red}{Red text}
\textcolor{blue}{Blue text}
\textcolor{green!50!black}{Dark green text}

% Color as declaration:
```

```latex
{\color{red} This entire block is red.}

% Define custom colors:
\definecolor{mycolor}{RGB}{102, 178, 255}
\textcolor{mycolor}{Custom colored text}

% Background color (requires xcolor with table option):
\colorbox{yellow}{Highlighted text}
\fcolorbox{red}{yellow}{Box with red border and yellow background
    }
```

Listing 2.9: Using colors

Examples: Red, Blue, Dark Green, Highlighted

## 2.6 Accents and Special Characters

| Accent | Command | Result | Alt (UTF-8) |
|---|---|---|---|
| Grave | `\‘{o}` | ò | ò |
| Acute | `\’{o}` | ó | ó |
| Circumflex | `\^{o}` | ô | ô |
| Tilde | `\~{o}` | õ | õ |
| Umlaut | `\"{o}` | ö | ö |
| Cedilla | `\c{c}` | ç | ç |
| Ogonek | `\k{a}` | ą | |
| Macron | `\={o}` | ō | |
| Breve | `\u{o}` | ŏ | |
| Dot | `\.{o}` | ȯ | |
| Ring | `\r{a}` | å | |
| Hacek | `\v{o}` | ǒ | |
| Double acute | `\H{o}` | ő | |

Table 2.4: Accent commands

## 2.7 Dashes and Quotes

| Type | Source | Result | Usage |
|---|---|---|---|
| Hyphen | `-` | mother-in-law | Compound words |
| En-dash | `--` | pages 1–10 | Ranges |
| Em-dash | `---` | Yes—indeed | Interruptions |
| Minus | `$-$` | − | Mathematical minus |

Table 2.5: Types of dashes

| **Type** | **Source** | **Result** |
|---|---|---|
| Opening double quotes | ` ` | “Hello” |
| Closing double quotes | ' ' | “Hello” |
| Opening single quote | ` | ‘Hello’ |
| Closing single quote | ' | ‘Hello’ |

Table 2.6: Quotation marks

*Note: LaTeX does not use regular keyboard quotes (").  Always use backticks for opening quotes.*

# Chapter 3

# Lists

## 3.1 Itemize (Bulleted Lists)

```latex
\begin{itemize}
    \item First item
    \item Second item
    \item Third item
\end{itemize}
```

Listing 3.1: Basic itemize list

Result:

- First item
- Second item
- Third item

### 3.1.1 Nested Itemize

```latex
\begin{itemize}
    \item First level
    \begin{itemize}
        \item Second level
        \begin{itemize}
            \item Third level
            \begin{itemize}
                \item Fourth level
            \end{itemize}
        \end{itemize}
    \end{itemize}
\end{itemize}
```

Listing 3.2: Nested itemize

Result:

- First level

- Second level
  - Third level
    - Fourth level

## 3.2 Enumerate (Numbered Lists)

```
\begin{enumerate}
    \item First item
    \item Second item
    \item Third item
\end{enumerate}
```

Listing 3.3: Basic enumerate list

Result:

1. First item
2. Second item
3. Third item

### 3.2.1 Nested Enumerate

1. First level
   (a) Second level
      i. Third level
         A. Fourth level
2. Back to first level

## 3.3 Description Lists

```
\begin{description}
    \item[Term 1] Definition or description of term 1.
    \item[Term 2] Definition or description of term 2.
    \item[Term 3] Definition or description of term 3.
\end{description}
```

Listing 3.4: Description list

Result:

**LaTeX** A document preparation system.

**TeX** The typesetting engine by Donald Knuth.

**PDF** Portable Document Format.

## 3.4 Customizing Lists with enumitem

The `enumitem` package provides powerful list customization.

### 3.4.1 Custom Labels

```latex
% Roman numerals:
\begin{enumerate}[label=\Roman*.]
    \item First
    \item Second
\end{enumerate}

% Alphabetical:
\begin{enumerate}[label=\alph*)]
    \item First
    \item Second
\end{enumerate}

% Custom format:
\begin{enumerate}[label=Step \arabic*:)]
    \item First
    \item Second
\end{enumerate}
```

Listing 3.5: Custom enumerate labels

Roman numerals:

I. First

II. Second

III. Third

Alphabetical:

a) First

b) Second

c) Third

Custom format:

Step 1: Gather requirements

Step 2: Design solution

Step 3: Implement

Step 4: Test

### 3.4.2 Custom Bullet Symbols

```latex
\begin{itemize}[label=$\star$]
    \item Star bullet
```

```latex
    \item Another star
\end{itemize}

\begin{itemize}[label=\textbullet]
    \item Standard bullet
\end{itemize}

\begin{itemize}[label=$\checkmark$]
    \item Checkmark bullet
\end{itemize}
```

Listing 3.6: Custom bullets

- ⋆ Star bullet one
- ⋆ Star bullet two
- ✓ Completed task one
- ✓ Completed task two

### 3.4.3 Spacing Control

```latex
\begin{itemize}[nosep]            % No extra spacing between items
    \item Compact item 1
    \item Compact item 2
\end{itemize}

\begin{itemize}[topsep=0pt]      % No space above/below list
    \item Item 1
    \item Item 2
\end{itemize}

\begin{enumerate}[start=5]       % Start counting from 5
    \item Fifth
    \item Sixth
\end{enumerate}
```

Listing 3.7: List spacing

Compact list (`nosep`):

- Compact item 1
- Compact item 2
- Compact item 3

Starting from 5:

5. Fifth
6. Sixth
7. Seventh

## 3.5 Inline Lists

```latex
\usepackage[inline]{enumitem}

The colors are:
\begin{enumerate*}[label=\arabic*.]
    \item red, \item green, and \item blue.
\end{enumerate*}
```

Listing 3.8: Inline list with enumitem

*Note: Inline lists require* `\usepackage[inline]{enumitem}`.

## 3.6 List Summary

| Environment | Best For | Default Label |
|---|---|---|
| `itemize` | Unordered items | Bullets (•, –, *, ·) |
| `enumerate` | Ordered/sequential | Numbers (1., (a), i., A.) |
| `description` | Definitions/terms | Bold label text |

Table 3.1: List environments summary

# Chapter 4

# Tables

## 4.1 Basic Table (tabular)

The `tabular` environment is the fundamental table environment in LaTeX.

### 4.1.1 Column Types

| Spec | Description |
|---|---|
| `l` | Left-aligned column |
| `c` | Centered column |
| `r` | Right-aligned column |
| `p{width}` | Paragraph column with fixed width |
| `\|` | Vertical line |
| `@{text}` | Replace inter-column space with *text* |

Table 4.1: Column type specifications

### 4.1.2 Basic Example

```latex
\begin{tabular}{|l|c|r|}
    \hline
    Left & Center & Right \\
    \hline
    A    & B      & C \\
    \hline
\end{tabular}
```

Listing 4.1: Basic tabular

Result:

| Left | Center | Right |
|---|---|---|
| A | B | C |

## 4.2 Table Syntax

```latex
% &       -- column separator
% \\      -- end of row
% \hline -- horizontal line (spans full width)
% \cline{2-3} -- partial horizontal line (columns 2 to 3)

\begin{tabular}{|l|c|r|}
    \hline
    Name  & Age & Score \\
    \hline
    Alice & 25  & 95 \\
    Bob   & 30  & 87 \\
    \cline{2-3}
    \multicolumn{1}{c|}{} & \multicolumn{2}{c}{Partial line} \\
    \hline
\end{tabular}
```

Listing 4.2: Table syntax reference

## 4.3 Professional Tables with booktabs

The `booktabs` package produces publication-quality tables without vertical lines.

```latex
\usepackage{booktabs}

\begin{tabular}{lcc}
    \toprule
    Name  & Age & Score \\
    \midrule
    Alice & 25  & 95 \\
    Bob   & 30  & 87 \\
    Carol & 28  & 92 \\
    \bottomrule
\end{tabular}
```

Listing 4.3: booktabs example

Result:

| **Name** | **Age** | **Score** |
|---|---|---|
| Alice | 25 | 95 |
| Bob | 30 | 87 |
| Carol | 28 | 92 |
| **Average** | **27.7** | **91.3** |

Table 4.2: Student scores using booktabs

| booktabs command | Purpose |
|---|---|
| `\toprule` | Top rule (thicker) |
| `\midrule` | Middle rule (medium) |
| `\bottomrule` | Bottom rule (thicker) |
| `\cmidrule{a-b}` | Partial rule (thinner) |
| `\addlinespace` | Extra vertical space |

Table 4.3: booktabs commands

## 4.4 Multi-column and Multi-row

### 4.4.1 Multi-column (\multicolumn)

```
\multicolumn{number_of_columns}{alignment}{text}
```

Listing 4.4: Multicolumn

| | Math | Physics | Chemistry |
|---|---|---|---|
| Alice | 95 | 88 | 91 |
| Bob | 87 | 92 | 85 |
| *Class Average* | | **89.7** | |

Table 4.4: Multicolumn example

### 4.4.2 Multi-row (multirow package)

```
\usepackage{multirow}
\multirow{number_of_rows}{width}{text}
```

Listing 4.5: Multirow

| Category | Subject | Students | Avg Score |
|---|---|---|---|
| Science | Physics | 45 | 88 |
| | Chemistry | 38 | 85 |
| Arts | History | 32 | 91 |
| | Literature | 29 | 93 |

Table 4.5: Multirow example

## 4.5 Fixed-width Tables (tabularx)

```
\usepackage{tabularx}

\begin{tabularx}{\textwidth}{lX}
```

```latex
    \toprule
    Name & Description \\
    \midrule
    Item 1 & This text automatically wraps to fill
             the available column width. \\
    Item 2 & Another long description that wraps
             automatically in the X column. \\
    \bottomrule
\end{tabularx}
```

Listing 4.6: tabularx

Result:

| Command | Description |
|---|---|
| \LaTeX | Produces the LaTeX logo with proper kerning and raised "A". |
| \TeX | Produces the TeX logo. |
| \today | Produces the current date. |
| \tableofcontents | Generates the table of contents from sectioning commands. |

Table 4.6: tabularx with auto-wrapping X column

## 4.6 Multi-page Tables (longtable)

```latex
\usepackage{longtable}

\begin{longtable}{lcc}
    \caption{A long table spanning multiple pages} \\
    \toprule
    Name & Value & Note \\
    \midrule
    \endfirsthead

    \toprule
    Name & Value & Note \\
    \midrule
    \endhead

    \midrule
    \multicolumn{3}{r}{\textit{Continued on next page}} \\
    \endfoot

    \bottomrule
    \endlastfoot

    Data1 & 100 & Note1 \\
    Data2 & 200 & Note2 \\
    % ... many more rows ...
```

```latex
\end{longtable}
```

Listing 4.7: longtable

## 4.7 Table Placement (Float)

Tables inside `table` environment are *floating bodies*:

```latex
\begin{table}[placement]
    % placement can be:
    % h = here        (approximately at current location)
    % t = top         (top of page)
    % b = bottom      (bottom of page)
    % p = page        (on a separate float page)
    % ! = override    (relax LaTeX's internal rules)
    % H = HERE        (requires float package -- exact placement)
    \centering
    \begin{tabular}{lc}
        ...
    \end{tabular}
    \caption{Table caption}
    \label{tab:mylabel}
\end{table}
```

Listing 4.8: Float placement specifiers

## 4.8 Colored Tables

```latex
\usepackage[table]{xcolor}

\rowcolor{gray!20}        % Color entire row
\cellcolor{blue!25}        % Color single cell
\columncolor{green!15}     % Color entire column (in column spec)
```

Listing 4.9: Colored table rows

| Header 1 | Header 2 | Header 3 |
|---|---|---|
| Row 1, Col 1 | Row 1, Col 2 | Row 1, Col 3 |
| Row 2, Col 1 | Row 2, Col 2 | Row 2, Col 3 |
| Row 3, Col 1 | Row 3, Col 2 | Row 3, Col 3 |
| Row 4, Col 1 | Row 4, Col 2 | Row 4, Col 3 |

Table 4.7: Alternating row colors

## 4.9 Table Quick Reference

| **Symbol/Command** | **Meaning** |
|---|---|
| `&` | Column separator |
| `\\` | Row terminator |
| `\hline` | Full horizontal line |
| `\cline{a-b}` | Partial horizontal line |
| `\vline` | Vertical line (inline) |
| `\multicolumn{n}{spec}{text}` | Span $n$ columns |
| `\multirow{n}{w}{text}` | Span $n$ rows |
| `\toprule` | booktabs top rule |
| `\midrule` | booktabs middle rule |
| `\bottomrule` | booktabs bottom rule |

Table 4.8: Table syntax quick reference

# Chapter 5

# Mathematics

LaTeX is renowned for its mathematical typesetting. This chapter covers everything from basic inline formulas to advanced constructs.

## 5.1 Math Modes

There are several ways to enter math mode:

| Type | Syntax | Usage |
|---|---|---|
| Inline math | `$...$` | $E = mc^2$ in text |
| Display math | `\[...\]` | Centered, numbered |
| Equation env | `\begin{equation}` | Centered, numbered |
| Align env | `\begin{align}` | Multi-line aligned |

Table 5.1: Math mode entry points

```
% Inline: the formula flows with text
The energy is $E = mc^2$ according to Einstein.

% Display: the formula is centered on its own line
\[ E = mc^2 \]

% Equation: like display but with a number
\begin{equation}
    E = mc^2 \label{eq:einstein}
\end{equation}

% Starred version: no number
\begin{equation*}
    E = mc^2
\end{equation*}
```

Listing 5.1: Math modes

## 5.2 Superscripts and Subscripts

```
$x^2$                % Superscript: x squared
$x_i$                % Subscript: x sub i
$x_i^2$              % Both: x sub i squared
$x_{i,j}$            % Multi-character subscript: need braces
$a^{b^c}$            % Nested superscript
$2^{10} = 1024$      % Powers
```

Listing 5.2: Superscripts and subscripts

Results: $x^2$, $x_i$, $x_i^2$, $x_{i,j}$, $a^{b^c}$, $2^{10} = 1024$

## 5.3 Fractions

```
$\frac{a}{b}$                                % Basic fraction
$\dfrac{a}{b}$                               % Display-style fraction
$\tfrac{a}{b}$                               % Text-style fraction
$\cfrac{1}{1+\cfrac{1}{2}}$                 % Continued fraction
$\binom{n}{k}$                               % Binomial coefficient
```

Listing 5.3: Fractions

Results: $\frac{a}{b}$, $\dfrac{a}{b}$, $\cfrac{1}{1+\cfrac{1}{2}}$, $\binom{n}{k}$

## 5.4 Roots

```
$\sqrt{x}$              % Square root
$\sqrt[3]{x}$           % Cube root
$\sqrt[n]{x}$           % nth root
$\sqrt{a^2 + b^2}$      % Complex expression
```

Listing 5.4: Roots

Results: $\sqrt{x}$, $\sqrt[3]{x}$, $\sqrt[n]{x}$, $\sqrt{a^2 + b^2}$

## 5.5 Sums, Products, Integrals

```
$\sum_{i=1}^{n} x_i$          % Summation
$\prod_{i=1}^{n} x_i$         % Product
$\int_{a}^{b} f(x) dx$        % Definite integral
$\iint_{D} f(x,y) dA$         % Double integral
$\iiint_{V} f(x,y,z) dV$      % Triple integral
$\oint_{C} F \cdot dr$        % Contour integral
$\lim_{x \to \infty} f(x)$   % Limit
$\sup_{x \in S} f(x)$        % Supremum
$\max_{x \in S} f(x)$        % Maximum
```

```
$\min_{x \in S} f(x)$          % Minimum
```

Listing 5.5: Big operators

Display style results:

$$\sum_{i=1}^{n} x_i \qquad \prod_{i=1}^{n} x_i \qquad \int_a^b f(x)\,dx \qquad \lim_{x\to\infty} f(x)$$

## 5.6 Greek Letters

| Command | Result | Command | Result |
|---|---|---|---|
| \alpha | $\alpha$ | \Alpha | $A$ |
| \beta | $\beta$ | \gamma | $\gamma$ |
| \delta | $\delta$ | \epsilon | $\epsilon$ |
| \varepsilon | $\varepsilon$ | \zeta | $\zeta$ |
| \eta | $\eta$ | \theta | $\theta$ |
| \iota | $\iota$ | \kappa | $\kappa$ |
| \lambda | $\lambda$ | \mu | $\mu$ |
| \nu | $\nu$ | \xi | $\xi$ |
| \pi | $\pi$ | \rho | $\rho$ |
| \sigma | $\sigma$ | \tau | $\tau$ |
| \upsilon | $\upsilon$ | \phi | $\phi$ |
| \varphi | $\varphi$ | \chi | $\chi$ |
| \psi | $\psi$ | \omega | $\omega$ |
| \Gamma | $\Gamma$ | \Delta | $\Delta$ |
| \Theta | $\Theta$ | \Lambda | $\Lambda$ |
| \Xi | $\Xi$ | \Pi | $\Pi$ |
| \Sigma | $\Sigma$ | \Phi | $\Phi$ |
| \Psi | $\Psi$ | \Omega | $\Omega$ |

Table 5.2: Greek letters

## 5.7 Mathematical Operators and Functions

```
$\sin x$      $\cos x$      $\tan x$
$\arcsin x$   $\arccos x$   $\arctan x$
$\sinh x$     $\cosh x$     $\tanh x$
$\ln x$       $\log x$      $\exp x$
$\det A$      $\dim V$      $\ker f$
$\deg f$      $\gcd(a,b)$   $\hom(A,B)$
$\inf S$      $\sup S$      $\max S$
$\min S$      $\lim f$      $\limsup f$
```

Listing 5.6: Math operators

## 5.8 Relations and Binary Operators

| Command | Symbol | Command | Symbol |
|---|---|---|---|
| `<` | $<$ | `>` | $>$ |
| `\leq` | $\leq$ | `\geq` | $\geq$ |
| `\neq` | $\neq$ | `\approx` | $\approx$ |
| `\equiv` | $\equiv$ | `\sim` | $\sim$ |
| `\propto` | $\propto$ | `\ll` | $\ll$ |
| `\gg` | $\gg$ | `\in` | $\in$ |
| `\subset` | $\subset$ | `\supset` | $\supset$ |
| `\subseteq` | $\subseteq$ | `\supseteq` | $\supseteq$ |
| `\cup` | $\cup$ | `\cap` | $\cap$ |
| `\setminus` | $\setminus$ | `\emptyset` | $\emptyset$ |
| `\pm` | $\pm$ | `\mp` | $\mp$ |
| `\times` | $\times$ | `\div` | $\div$ |
| `\cdot` | $\cdot$ | `\circ` | $\circ$ |
| `\otimes` | $\otimes$ | `\oplus` | $\oplus$ |

Table 5.3: Common mathematical symbols

## 5.9 Arrows

| Command | Arrow | Command | Arrow |
|---|---|---|---|
| `\leftarrow` | $\leftarrow$ | `\rightarrow` | $\rightarrow$ |
| `\Leftarrow` | $\Leftarrow$ | `\Rightarrow` | $\Rightarrow$ |
| `\leftrightarrow` | $\leftrightarrow$ | `\Leftrightarrow` | $\Leftrightarrow$ |
| `\mapsto` | $\mapsto$ | `\longmapsto` | $\longmapsto$ |
| `\uparrow` | $\uparrow$ | `\downarrow` | $\downarrow$ |
| `\nearrow` | $\nearrow$ | `\searrow` | $\searrow$ |

Table 5.4: Arrow symbols

## 5.10 Matrices

The `amsmath` package provides several matrix environments:

```latex
% Parentheses: ( )
\begin{pmatrix}
    a & b \\
    c & d
\end{pmatrix}

% Brackets: [ ]
\begin{bmatrix}
    a & b \\
```

```latex
    c & d
\end{bmatrix}

% Braces: { }
\begin{Bmatrix}
    a & b \\
    c & d
\end{Bmatrix}

% Vertical bars: | |
\begin{vmatrix}
    a & b \\
    c & d
\end{vmatrix}

% Double vertical bars: || ||
\begin{Vmatrix}
    a & b \\
    c & d
\end{Vmatrix}

% No delimiters
\begin{matrix}
    a & b \\
    c & d
\end{matrix}

% With dots:
\begin{pmatrix}
    a_{11} & a_{12} & \cdots & a_{1n} \\
    a_{21} & a_{22} & \cdots & a_{2n} \\
    \vdots & \vdots & \ddots & \vdots \\
    a_{m1} & a_{m2} & \cdots & a_{mn}
\end{pmatrix}
```

Listing 5.7: Matrices

Results:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

## 5.11 Multi-line Equations (align)

```latex
\begin{align}
    f(x) &= x^2 + 2x + 1 \\
```

```
            &= (x+1)^2 \\
            &= x^2 + 2x + 1 \label{eq:expand}
\end{align}

% align without numbering:
\begin{align*}
    f(x) &= x^2 + 2x + 1 \\
            &= (x+1)^2
\end{align*}

% Number specific lines only:
\begin{align}
    f(x) &= x^2 + 2x + 1 \nonumber \\
            &= (x+1)^2 \\
            &= x^2 + 2x + 1 \nonumber
\end{align}

% Multiple alignment columns:
\begin{align*}
    x &= 1  &  y &= 2  &  z &= 3 \\
    x' &= 4 &  y' &= 5 &  z' &= 6
\end{align*}
```

Listing 5.8: align environment

Result:

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x+1)^2 \end{aligned} \tag{5.1}$$

## 5.12 Cases and Piecewise Functions

```
\begin{equation}
    |x| =
    \begin{cases}
        x  & \text{if } x \geq 0 \\
        -x & \text{if } x < 0
    \end{cases}
\end{equation}
```

Listing 5.9: cases environment

Result:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \tag{5.2}$$

## 5.13 Accents in Math Mode

| Command | Result | Name |
|---|---|---|
| `\hat{a}` | $\hat{a}$ | Hat |
| `\bar{a}` | $\bar{a}$ | Bar |
| `\tilde{a}` | $\tilde{a}$ | Tilde |
| `\vec{a}` | $\vec{a}$ | Vector arrow |
| `\dot{a}` | $\dot{a}$ | Dot |
| `\ddot{a}` | $\ddot{a}$ | Double dot |
| `\vec{a}` | $\vec{a}$ | Vector |
| `\widehat{abc}` | $\widehat{abc}$ | Wide hat |
| `\widetilde{abc}` | $\widetilde{abc}$ | Wide tilde |
| `\overline{abc}` | $\overline{abc}$ | Overline |
| `\underline{abc}` | $\underline{abc}$ | Underline |
| `\overbrace{abc}` | $\overbrace{abc}$ | Overbrace |
| `\underbrace{abc}` | $\underbrace{abc}$ | Underbrace |

Table 5.5: Math accents

## 5.14 Spacing in Math Mode

| Command | Space | Example |
|---|---|---|
| `\,` | Thin space | $a\,b$ |
| `\:` | Medium space | $a\:b$ |
| `\;` | Thick space | $a\;b$ |
| `\!` | Negative thin | $a\!b$ |
| `\quad` | 1em space | $a\quad b$ |
| `\qquad` | 2em space | $a\qquad b$ |
| `\,` | | $ab$ (no space for comparison) |

Table 5.6: Math spacing commands

## 5.15 Text Inside Math Mode

```
$\text{for all } x > 0$          % Regular text in math
$\text{if and only if}$           % Using amsmath \text
$\mathrm{d}x$                     % Upright d for differentials
$\text{Area} = \pi r^2$           % Upright labels
$\text{let } n = 5$               % Mixed text and math
```

Listing 5.10: Text in math

## 5.16 Theorems and Proofs

```latex
% In preamble (amsthm package):
\newtheorem{theorem}{Theorem}[section]
\newtheorem{lemma}[theorem]{Lemma}
\newtheorem{corollary}[theorem]{Corollary}
\theoremstyle{definition}
\newtheorem{definition}{Definition}[section]
\theoremstyle{remark}
\newtheorem{remark}{Remark}

% In document:
\begin{theorem}[Pythagorean]
For a right triangle: $a^2 + b^2 = c^2$.
\end{theorem}

\begin{proof}
By construction of squares on each side...
\end{proof}
```

Listing 5.11: Theorem environments

**Theorem 5.16.1** (Pythagorean)**.** *For a right triangle with legs $a$, $b$ and hypotenuse $c$:*

$$a^2 + b^2 = c^2$$

*Proof.* Left as an exercise for the reader. □

**Definition 5.16.1.** A **prime number** is a natural number greater than 1 that has no positive divisors other than 1 and itself.

## 5.17 Comprehensive Math Example

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi} \tag{5.3}$$

$$e^{i\pi} + 1 = 0 \tag{5.4}$$

The Gaussian integral (5.3) and Euler's identity (5.4) are two of the most beautiful equations in mathematics.

# Chapter 6

# Figures & Images

## 6.1 Including Images (graphicx)

```latex
\usepackage{graphicx}

% Basic image:
\includegraphics{filename}

% With options:
\includegraphics[width=5cm]{filename}
\includegraphics[height=3cm]{filename}
\includegraphics[scale=0.5]{filename}
\includegraphics[width=\textwidth]{filename}
\includegraphics[width=0.8\textwidth]{filename}
\includegraphics[angle=90]{filename}        % Rotate 90 degrees
\includegraphics[clip=true, trim=1cm 2cm 1cm 2cm]{filename} % Crop

% Supported formats: PDF, PNG, JPG, EPS
```

Listing 6.1: Including images

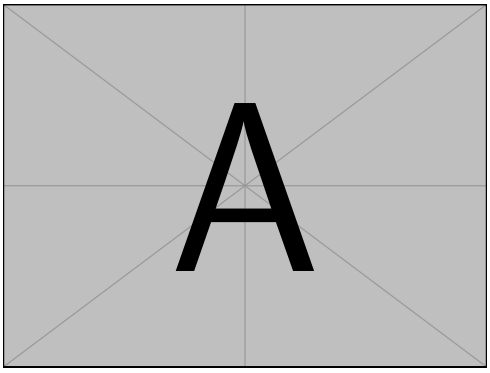



Figure 6.1: Example image A using `example-image-a`

## 6.2 Figure Environment (Float)

```latex
\begin{figure}[htbp]
    \centering
    \includegraphics[width=0.6\textwidth]{myimage}
    \caption{Description of the image.}
    \label{fig:mylabel}
\end{figure}

% Placement specifiers:
% h = here, t = top, b = bottom, p = float page
% ! = override LaTeX rules, H = exactly HERE (float package)
```

Listing 6.2: Figure float

### 6.2.1 Referencing Figures

As seen in Figure 6.1, we can reference images by their label.

```latex
% In text:
As shown in Figure~\ref{fig:mylabel} ...

% The ~ prevents a line break between "Figure" and the number.
```

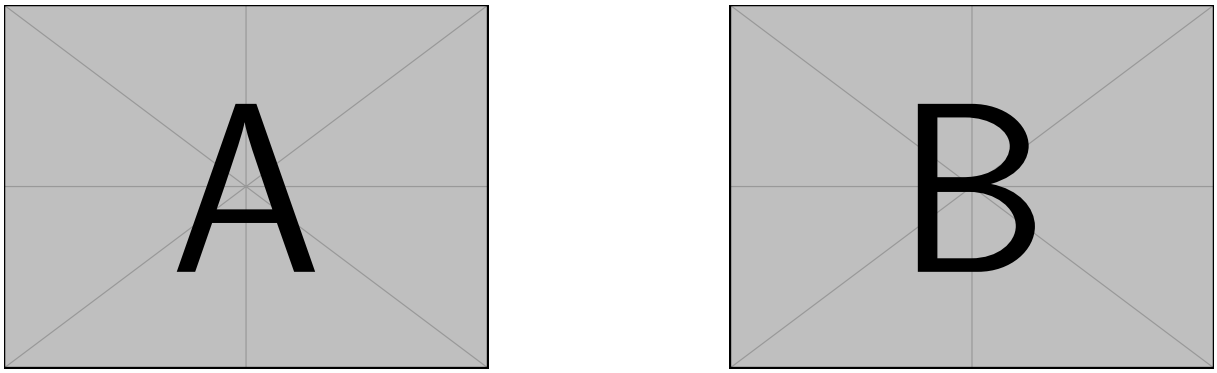
Listing 6.3: Referencing figures

## 6.3 Subfigures (subcaption package)

```latex
\usepackage{subcaption}

\begin{figure}[htbp]
    \centering
    \begin{subfigure}[b]{0.45\textwidth}
        \includegraphics[width=\textwidth]{image1}
        \caption{First subfigure}
        \label{fig:sub1}
    \end{subfigure}
    \hfill
    \begin{subfigure}[b]{0.45\textwidth}
        \includegraphics[width=\textwidth]{image2}
        \caption{Second subfigure}
        \label{fig:sub2}
    \end{subfigure}
    \caption{Overall caption for both subfigures}
    \label{fig:combined}
\end{figure}
```

Listing 6.4: Subfigures




(a) Subfigure A (b) Subfigure B

Figure 6.2: Two subfigures side by side

## 6.4 Image Size Options

| Option | Description |
|---|---|
| `width=5cm` | Set exact width |
| `height=3cm` | Set exact height |
| `width=\textwidth` | Scale to text width |
| `width=0.5\textwidth` | Half text width |
| `scale=0.5` | Scale to 50% |
| `angle=45` | Rotate 45 degrees |
| `clip=true` | Clip to bounding box |
| `trim=l b r t` | Trim from left, bottom, right, top |
| `keepaspectratio` | Maintain proportions |

Table 6.1: Image sizing options

## 6.5 Wrapping Text Around Images

```latex
\usepackage{wrapfig}

\begin{wrapfigure}{r}{0.4\textwidth}
    \centering
    \includegraphics[width=\linewidth]{image}
    \caption{Wrapped figure}
\end{wrapfigure}

This text will wrap around the figure on the left side.
% {r} = right side, {l} = left side, {R} = right (allows floating
    )
```

Listing 6.5: wrapfig

## 6.6 Drawing Simple Graphics with TikZ (Preview)

TikZ can also be used inside figure environments:

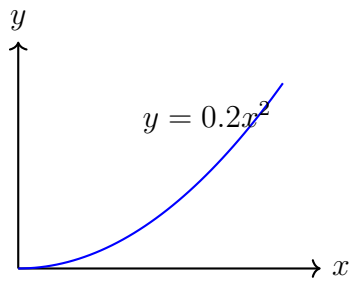



Figure 6.3: A simple plot drawn with TikZ

# Chapter 7

# Cross References & Hyperlinks

## 7.1 Labels and References

LaTeX provides a powerful cross-referencing system. Every numbered element can be labeled and referenced.

```latex
% Step 1: Place a label after the numbered element
\section{Introduction}
\label{sec:intro}

\begin{equation}
    E = mc^2
    \label{eq:einstein}
\end{equation}

\begin{figure}[htbp]
    \caption{My figure}
    \label{fig:myfig}
\end{figure}

\begin{table}[htbp]
    \caption{My table}
    \label{tab:mytab}
\end{table}

% Step 2: Reference it in text
As discussed in Section~\ref{sec:intro}...
Equation~\ref{eq:einstein} shows...
See Figure~\ref{fig:myfig} and Table~\ref{tab:mytab}.
```

Listing 7.1: Label and reference

### 7.1.1 Label Naming Conventions

| Element | Prefix | Example |
|---|---|---|
| Chapter | `ch:` | `ch:intro` |
| Section | `sec:` | `sec:methods` |
| Subsection | `ssec:` | `ssec:data` |
| Figure | `fig:` | `fig:diagram` |
| Table | `tab:` | `tab:results` |
| Equation | `eq:` | `eq:euler` |
| Listing | `lst:` | `lst:python` |
| Item | `itm:` | `itm:first` |

Table 7.1: Recommended label prefixes

### 7.1.2 Pageref and Nameref

```
% Reference the page number:
See page~\pageref{sec:intro}.

% Reference the title/name:
Section~\nameref{sec:intro} discusses...

% Reference both name and page with autoref (hyperref):
\autoref{sec:intro}  % Produces "Section 1"
\autoref{fig:myfig}  % Produces "Figure 1"
\autoref{eq:einstein}% Produces "Equation 1"
```

Listing 7.2: Other reference commands

## 7.2 Hyperlinks (hyperref package)

```
\usepackage{hyperref}

% URL links:
\url{https://www.latex-project.org/}
\href{https://www.latex-project.org/}{LaTeX Project}

% Email link:
\href{mailto:user@example.com}{user@example.com}

% File link:
\href{file:///path/to/file}{Open file}

% Internal link (jump to label):
\hyperref[sec:intro]{Go to Introduction}

% Configuration:
\hypersetup{
```

```
    colorlinks=true,        % Colored links (not boxed)
    linkcolor=blue,         % Internal link color
    filecolor=magenta,      % File link color
    urlcolor=cyan,          % URL color
    citecolor=green,        % Citation color
}
```

Listing 7.3: Hyperlinks

Examples:

- URL: https://www.latex-project.org/
- Hyperlinked text: Overleaf — Online LaTeX Editor

## 7.3 Table of Contents

```
% Basic table of contents:
\tableofcontents

% List of figures:
\listoffigures

% List of tables:
\listoftables

% Limit TOC depth:
\setcounter{tocdepth}{2}  % Show up to subsection

% Set sectioning number depth:
\setcounter{secnumdepth}{3}  % Number up to subsubsection

% Add unnumbered section to TOC:
\section*{Acknowledgments}
\addcontentsline{toc}{section}{Acknowledgments}
```

Listing 7.4: Table of contents

### 7.3.1 Custom TOC Entry

```
% Add arbitrary entries to TOC:
\addcontentsline{toc}{chapter}{Preface}

% For the list of figures/tables:
\addcontentsline{lof}{figure}{My Figure Title}
```

Listing 7.5: Custom TOC entries

# Chapter 8

# Bibliography & Citations

## 8.1 Basic Bibliography with thebibliography

The simplest way to create a bibliography is using the `thebibliography` environment directly.

```latex
% At the end of your document (before \end{document}):
\begin{thebibliography}{9}
    \bibitem{knuth1984}
        Donald E. Knuth,
        \textit{The TeXbook},
        Addison-Wesley, 1984.

    \bibitem{lamport1994}
        Leslie Lamport,
        \textit{LaTeX: A Document Preparation System},
        Addison-Wesley, 2nd edition, 1994.

    \bibitem{mittelbach2004}
        Frank Mittelbach and Michel Goossens,
        \textit{The LaTeX Companion},
        Addison-Wesley, 2nd edition, 2004.
\end{thebibliography}

% Cite in text:
Knuth wrote the original TeX system~\cite{knuth1984}.
LaTeX was created by Lamport~\cite{lamport1994}.
```

Listing 8.1: thebibliography environment

### 8.1.1 Citation Commands

| Command | Output | Description |
|---|---|---|
| `\cite{key}` | [1] | Basic citation |
| `\cite[p.42]{key}` | [1, p.42] | With page number |
| `\cite{a,b,c}` | [1,2,3] | Multiple citations |

Table 8.1: Basic citation commands

## 8.2 BibTeX with .bib Files

For larger projects, use a `.bib` file to manage references.

### 8.2.1 The .bib File

```
@article{einstein1905,
    author  = {Albert Einstein},
    title   = {On the Electrodynamics of Moving Bodies},
    journal = {Annalen der Physik},
    year    = {1905},
    volume  = {322},
    number  = {10},
    pages   = {891--921},
    doi     = {10.1002/andp.19053221004}
}

@book{knuth1984,
    author    = {Donald E. Knuth},
    title     = {The TeXbook},
    publisher = {Addison-Wesley},
    year      = {1984},
    isbn      = {978-0201134483}
}

@inproceedings{lamport1998,
    author    = {Leslie Lamport},
    title     = {The Part-Time Parliament},
    booktitle = {ACM Transactions on Computer Systems},
    year      = {1998},
    volume    = {16},
    number    = {2},
    pages     = {133--169}
}

@online{overleaf2024,
    author  = {{Overleaf}},
    title   = {Documentation},
    year    = {2024},
    url     = {https://www.overleaf.com/learn}
```

```
}

@phdthesis{smith2020,
    author = {Jane Smith},
    title  = {Advanced Topics in Computer Science},
    school = {MIT},
    year   = {2020}
}
```

Listing 8.2: Sample .bib file (references.bib)

### 8.2.2 Entry Types

| Type | Used For |
|---|---|
| `@article` | Journal articles |
| `@book` | Books |
| `@booklet` | Booklets / pamphlets |
| `@inbook` | A chapter or section in a book |
| `@incollection` | A chapter in a collected works |
| `@inproceedings` | Conference papers |
| `@manual` | Technical documentation |
| `@mastersthesis` | Master's thesis |
| `@misc` | Miscellaneous |
| `@phdthesis` | PhD thesis |
| `@proceedings` | Conference proceedings |
| `@techreport` | Technical reports |
| `@unpublished` | Unpublished works |
| `@online` | Web resources |

Table 8.2: BibTeX entry types

### 8.2.3 Common Fields

| Field | Required | Description |
|---|---|---|
| `author` | Most entries | Author names |
| `title` | Most entries | Title of work |
| `year` | Most entries | Year of publication |
| `publisher` | book, inbook | Publisher name |
| `journal` | article | Journal name |
| `volume` | article | Volume number |
| `number` | article | Issue number |
| `pages` | article | Page range |
| `doi` | Optional | Digital Object Identifier |
| `url` | Optional | URL to resource |
| `isbn` | Optional | ISBN number |
| `abstract` | Optional | Abstract text |
| `keywords` | Optional | Keywords |

Table 8.3: Common BibTeX fields

## 8.3 Using BibTeX

```latex
% In your .tex file, at the end:
\bibliographystyle{plain}     % Choose a style
\bibliography{references}     % Name of .bib file (without .bib)

% Compilation steps:
% 1. pdflatex main.tex       (first pass)
% 2. bibtex main              (process citations)
% 3. pdflatex main.tex       (second pass)
% 4. pdflatex main.tex       (third pass for references)
```

Listing 8.3: Using BibTeX

### 8.3.1 Bibliography Styles

| Style | Description |
|---|---|
| `plain` | Alphabetical by author, numbered [1] |
| `unsrt` | Same as plain, but in order of citation |
| `alpha` | Labels like [Knu84] (author-year abbrev.) |
| `abbrv` | Like plain, with abbreviated first names |
| `ieeetr` | IEEE Transactions style |
| `acm` | ACM style |
| `apalike` | APA-like style (author, year) |

Table 8.4: Common bibliography styles

## 8.4 Biblatex (Modern Approach)

```latex
% In preamble:
\usepackage[backend=biber, style=numeric]{biblatex}
\addbibresource{references.bib}

% In document body:
As shown by~\textcite{einstein1905}...
According to~\autocite{knuth1984}...

% At end of document:
\printbibliography

% Or with a title:
\printbibliography[title={Works Cited}]

% Or filter by type:
\printbibliography[type=article, title={Journal Articles}]
\printbibliography[type=book, title={Books}]

% Citation commands unique to biblatex:
\cite{key}            % [1]
\parencite{key}       % (Author, Year)
\textcite{key}        % Author (Year)
\autocite{key}        % Depends on style
\footcite{key}        % Footnote citation
```

Listing 8.4: biblatex package

## 8.5 Example Bibliography

Here is a working example with inline references:

# Bibliography


[1] Donald E. Knuth, *The TEXbook*, Addison-Wesley, 1984.

[2] Leslie Lamport, *LATEX: A Document Preparation System*, Addison-Wesley, 2nd edition, 1994.

[3] Frank Mittelbach and Michel Goossens, *The LATEX Companion*, Addison-Wesley, 2nd edition, 2004.

[4] Overleaf, *Learn LATEX in 30 Minutes*, https://www.overleaf.com/learn, 2024.

# Chapter 9

# Code Listings & Verbatim Text

## 9.1 Verbatim Text

The `verbatim` environment prints text exactly as typed, without interpreting any LaTeX commands.

```latex
% Inline verbatim:
\verb|This is \verbatim{test} text|

% Display verbatim:
\begin{verbatim}
    \textbf{This text is NOT bold}
    $E = mc^2$ is NOT an equation
    Everything appears exactly as typed!
\end{verbatim}

% Verbatim with different delimiter:
\verb!Special chars: $ % & # _ { }!

% With fancyvrb package (more options):
\usepackage{fancyvrb}
\begin{Verbatim}[fontsize=\small, frame=single, framesep=2mm]
    Custom verbatim with frame and smaller font
\end{Verbatim}
```

Listing 9.1: Verbatim environments

### 9.1.1 Inline Verbatim Examples

- `\documentclass{article}` the document class command
- `\usepackage{graphicx}` loading a package
- `$E = mc^2$` inline math

## 9.2 Code Listings (listings package)

The `listings` package provides sophisticated source code formatting.

### 9.2.1 Basic Listing

```python
def hello_world():
    print("Hello, World!")
```

Listing 9.2: Basic code listing

### 9.2.2 Listing with Language

```python
# Python example: Fibonacci sequence
def fibonacci(n):
    """Calculate the nth Fibonacci number."""
    if n <= 1:
        return n
    a, b = 0, 1
    for _ in range(2, n + 1):
        a, b = b, a + b
    return b

# Generate first 10 Fibonacci numbers
for i in range(10):
    print(f"F({i}) = {fibonacci(i)}")
```

Listing 9.3: Python example with syntax highlighting

### 9.2.3 Java Example

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");

        // Loop example
        for (int i = 0; i < 5; i++) {
            System.out.println("Iteration: " + i);
        }
    }
}
```

Listing 9.4: Java example

### 9.2.4 C/C++ Example

```cpp
#include <iostream>
#include <vector>
#include <algorithm>

```

```cpp
int main() {
    std::vector<int> numbers = {5, 2, 8, 1, 9, 3};

    // Sort ascending
    std::sort(numbers.begin(), numbers.end());

    // Print sorted vector
    for (const auto& n : numbers) {
        std::cout << n << " ";
    }
    std::cout << std::endl;

    return 0;
}
```

Listing 9.5: C++ example

### 9.2.5 JavaScript Example

```javascript
// Asynchronous data fetching
async function fetchData(url) {
    try {
        const response = await fetch(url);
        const data = await response.json();
        return data;
    } catch (error) {
        console.error("Error:", error);
    }
}
```

Listing 9.6: JavaScript example

## 9.3 Listing Styles and Configuration

```latex
% Global settings in preamble:
\lstset{
    basicstyle=\ttfamily\small,            % Base font
    keywordstyle=\color{blue}\bfseries,    % Keywords
    commentstyle=\color{green!50!black}\itshape, % Comments
    stringstyle=\color{red!70!black},      % Strings
    numbers=left,                          % Line numbers
    numberstyle=\tiny\color{gray},         % Line number style
    stepnumber=1,                          % Line number step
    numbersep=5pt,                         % Distance of numbers
    backgroundcolor=\color{gray!5},        % Background color
    frame=single,                          % Border frame
    rulecolor=\color{gray!50},             % Frame color
    breaklines=true,                       % Auto line break
    breakatwhitespace=false,               % Break only at
       whitespace
```

```
    tabsize=4,                          % Tab size
    showstringspaces=false,             % Don't show string
       spaces
    captionpos=b,                       % Caption at bottom
    title=\lstname,                     % Show filename as title
}
```

Listing 9.7: Listing configuration

*Note: Per-listing options go in square brackets:* `\begin{lstlisting}[language=Python, caption={...}]`.

## 9.4 Inline Code

```
% Use \lstinline for inline code:
The function \lstinline|print("hello")| outputs text.

% Short form (define a shorthand):
\lstMakeShortInline[language=Python]{|}
Use |print("hello")| in your code.
```

Listing 9.8: Inline code

## 9.5 Supported Languages

The listings package supports many programming languages:

| Language | Keyword | Language | Keyword |
|---|---|---|---|
| Python | `Python` | Java | `Java` |
| C | `C` | C++ | `C++` |
| JavaScript | `Java` | HTML | `HTML` |
| SQL | `SQL` | R | `R` |
| MATLAB | `Matlab` | Fortran | `Fortran` |
| Bash | `bash` | XML | `XML` |
| LaTeX | `[LaTeX]TeX` | Make | `make` |

Table 9.1: Some supported languages in the listings package

# Chapter 10

# TikZ Graphics

TikZ ("TikZ ist kein Zeichenprogramm") is a powerful package for creating graphics directly in LaTeX.

## 10.1 Basic Shapes

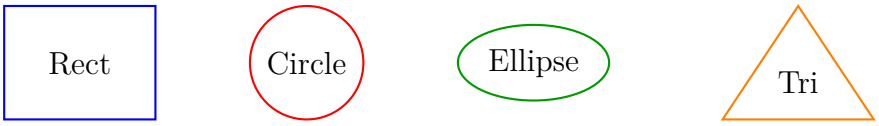



Figure 10.1: Basic shapes in TikZ

```
\begin{tikzpicture}
    \draw (0,0) rectangle (2,1);          % Rectangle
    \draw (4,1) circle (1);                % Circle
    \draw (7,1) ellipse (1.5 and 0.5);    % Ellipse
    \draw (0,0) -- (2,0) -- (1,2) -- cycle;  % Triangle
\end{tikzpicture}
```

Listing 10.1: Basic shapes

## 10.2 Lines and Paths

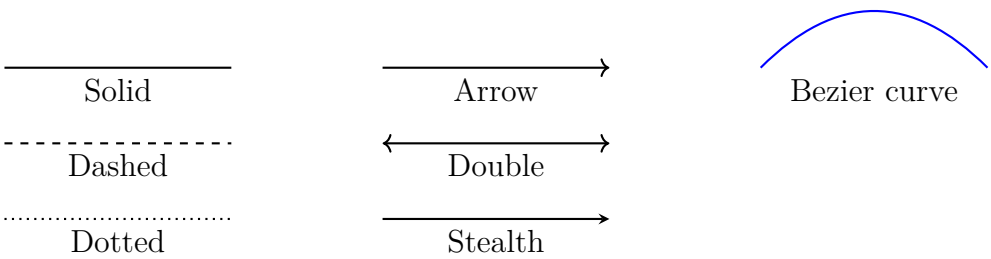



Figure 10.2: Lines and paths in TikZ

## 10.3 Filling and Colors

Figure 10.3: Filling and colors in TikZ

## 10.4 Nodes and Labels

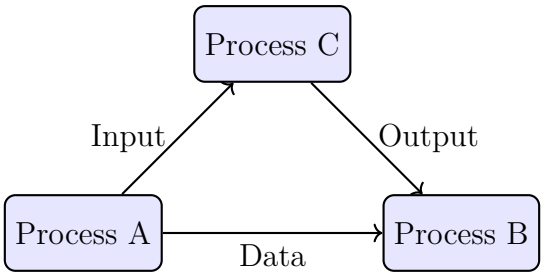



Figure 10.4: Nodes and connections in TikZ

```
\begin{tikzpicture}[
    box/.style={rectangle, draw, thick, fill=blue!10,
                minimum width=2cm, minimum height=1cm}
]
    \node[box] (a) at (0,0) {Node A};
    \node[box] (b) at (5,0) {Node B};

    \draw[->, thick] (a) -- (b) node[midway, above] {Label};
    \draw[->, thick, bend left=30] (a) to (b);
    \draw[->, thick, bend right=30] (a) to[bend right] (b);
\end{tikzpicture}
```

Listing 10.2: Nodes and edges

## 10.5 Flowchart

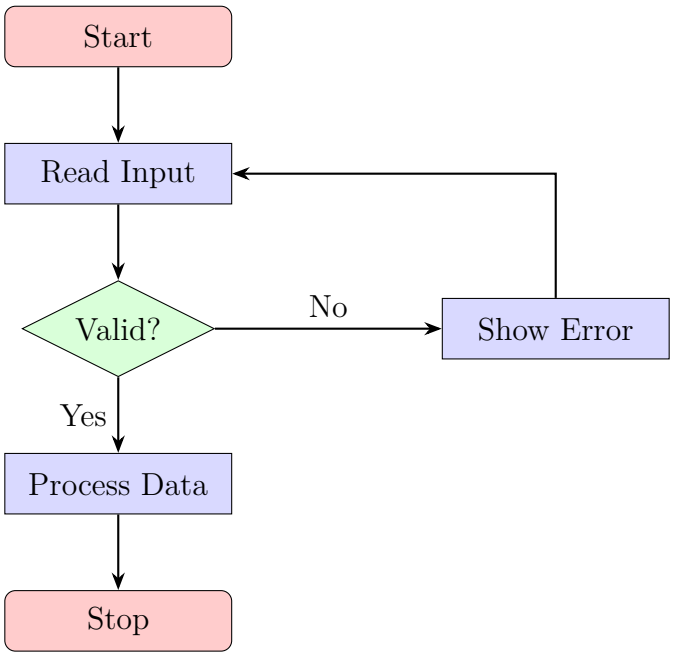



Figure 10.5: Flowchart using TikZ (requires `shapes.geometric` and `positioning` libraries)

## 10.6 Plots with pgfplots

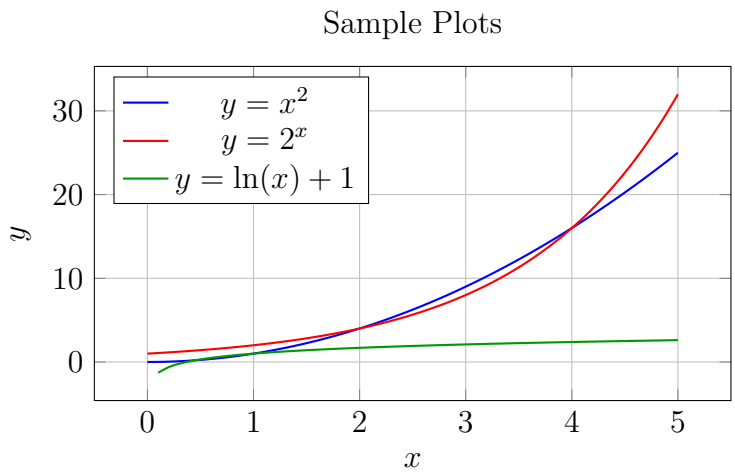



Figure 10.6: Function plots using pgfplots

```
\usepackage{pgfplots}
\pgfplotsset{compat=1.18}

\begin{tikzpicture}
    \begin{axis}[
        xlabel={$x$}, ylabel={$y$},
```

```
        legend pos=north west,
        grid=major,
    ]
        \addplot[blue, thick, domain=0:5, samples=50] {x^2};
        \addlegendentry{$y = x^2$}

        \addplot[red, thick, domain=0:5] {2^x};
        \addlegendentry{$y = 2^x$}
    \end{axis}
\end{tikzpicture}
```

Listing 10.3: pgfplots example

## 10.7 Bar Chart

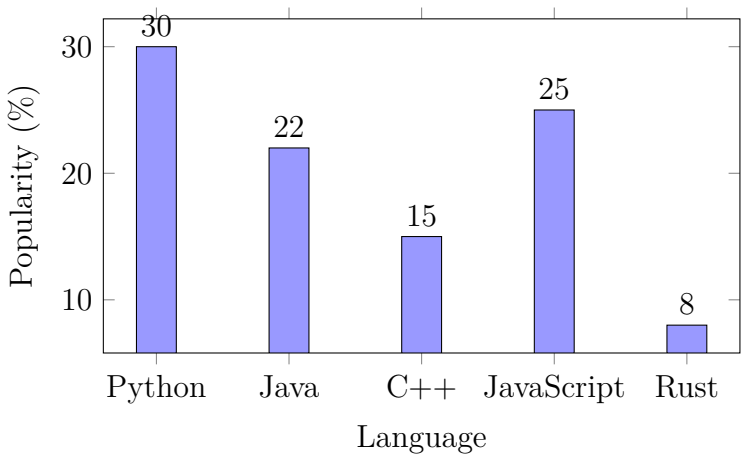



Figure 10.7: Bar chart using pgfplots

## 10.8 TikZ Styles

```
% Define reusable styles:
\tikzset{
    mybox/.style={
        rectangle, draw, thick,
        fill=blue!10,
        minimum width=2cm,
        minimum height=1cm,
        rounded corners=3pt,
        font=\sffamily
    },
    myarrow/.style={
        ->, >=stealth, thick, shorten >=1pt
    },
}

% Use them:
```

```
\node[mybox] (a) at (0,0) {Styled};
\node[mybox, fill=red!10] (b) at (3,0) {Override};
\draw[myarrow] (a) -- (b);
```

Listing 10.4: TikZ styles

# Chapter 11

# Page Layout & Headers/Footers

## 11.1 Page Geometry (geometry package)

```latex
\usepackage{geometry}

% Set all margins at once:
\geometry{margin=2.5cm}

% Set individual margins:
\geometry{
    left=3cm,
    right=3cm,
    top=2cm,
    bottom=2cm,
    bindingoffset=5mm   % Extra for binding
}

% Landscape orientation:
\geometry{landscape}

% Paper size:
\geometry{a4paper}     % or letterpaper, legalpaper, a5paper
```

Listing 11.1: Page layout

### 11.1.1 Understanding Page Layout

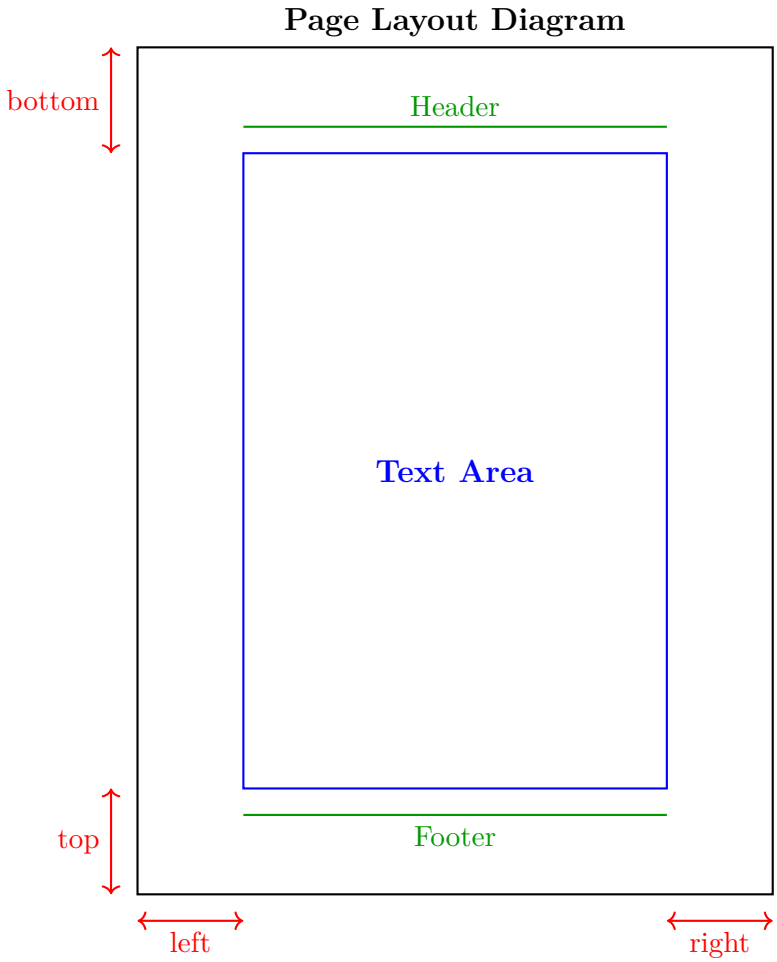



Figure 11.1: Page layout anatomy

## 11.2 Headers and Footers (fancyhdr)

```latex
\usepackage{fancyhdr}
\pagestyle{fancy}

% Clear all defaults:
\fancyhf{}

% Set header:
\fancyhead[L]{Left header}        % L = Left
\fancyhead[C]{Center header}      % C = Center
\fancyhead[R]{Right header}       % R = Right

% Set footer:
\fancyfoot[L]{Author Name}
\fancyfoot[C]{\thepage}           % Page number
\fancyfoot[R]{\today}

% For even/odd pages (twoside):
```

```latex
\fancyhead[LE,RO]{\thepage}       % Left Even, Right Odd
\fancyhead[RE,LO]{\leftmark}      % Right Even, Left Odd

% Rule thickness:
\renewcommand{\headrulewidth}{0.4pt}
\renewcommand{\footrulewidth}{0pt}  % No footer rule

% Chapter page style (plain):
\fancypagestyle{plain}{
    \fancyhf{}
    \fancyfoot[C]{\thepage}
    \renewcommand{\headrulewidth}{0pt}
}
```

Listing 11.2: fancyhdr package

### 11.2.1 Position Codes

| Code | Position |
|---|---|
| L | Left |
| C | Center |
| R | Right |
| E | Even page |
| O | Odd page |
| LE | Left on even page |
| RO | Right on odd page |

Table 11.1: Header/footer position codes

## 11.3 Page Numbering

```latex
% Numbering styles:
\pagenumbering{arabic}   % 1, 2, 3 (default)
\pagenumbering{roman}    % i, ii, iii
\pagenumbering{Roman}    % I, II, III
\pagenumbering{alph}     % a, b, c
\pagenumbering{Alph}     % A, B, C

% Set page counter:
\setcounter{page}{1}     % Start counting from 1

% No page number on current page:
\thispagestyle{empty}
```

Listing 11.3: Page numbering

## 11.4 Multiple Columns

```latex
% Document-wide twocolumn option:
\documentclass[twocolumn]{article}

% Switch mid-document:
\onecolumn    % Switch to one column
\twocolumn    % Switch to two columns

% With multicol package (more flexible):
\usepackage{multicol}

\begin{multicols}{3}  % 3 columns
    Text in column one continues into
    column two and then column three.
\end{multicols}

\begin{multicols}{2}[Section Title]
    Two-column section with a title.
\end{multicols}
```

Listing 11.4: Multiple columns

## 11.5 Page Breaks

```latex
\newpage          % Start a new page
\clearpage        % Flush all floats, then new page
\cleardoublepage  % For twoside: go to next odd-numbered page
\pagebreak        % Suggest a page break (stretchable)
\nopagebreak      % Suggest no page break
\linebreak        % Suggest a line break
\nolinebreak      % Suggest no line break
```

Listing 11.5: Page breaks

## 11.6 Paragraph Control

```latex
% Paragraph indent:
\setlength{\parindent}{1cm}          % Set indent width
\noindent                            % No indent for this paragraph
\indent                              % Force indent

% Paragraph skip (no indent, vertical space instead):
\usepackage{parskip}                 % Removes indent, adds space

% Or manually:
\setlength{\parindent}{0pt}
\setlength{\parskip}{6pt plus 2pt minus 1pt}
```

Listing 11.6: Paragraph control

# Chapter 12

# Floating Bodies & Captions

## 12.1 What Are Floats?

Floats are objects (figures, tables) that LaTeX places at optimal positions in the document, rather than exactly where they appear in the source code.

| Environment | Content |
|---|---|
| `figure` | Images, diagrams |
| `table` | Tables |
| `algorithm` | Algorithms (with algorithm package) |
| `listing` | Code listings (with listings package) |

Table 12.1: Float environments

## 12.2 Placement Specifiers

| Specifier | Meaning | Notes |
|---|---|---|
| `h` | Here | Approximately at the current location |
| `t` | Top | At the top of the page |
| `b` | Bottom | At the bottom of the page |
| `p` | Page | On a separate float page |
| `!` | Override | Override LaTeX's internal placement rules |
| `H` | HERE | Exactly here (requires `float` package) |

Table 12.2: Float placement specifiers

```latex
\begin{figure}[htbp]      % Try here, top, bottom, then page
    ...
\end{figure}

\begin{figure}[H]         % Exactly here (float package)
    ...
```

```latex
\end{figure}

\begin{table}[!ht]        % Override rules, try here then top
    ...
\end{table}
```

Listing 12.1: Float placement

## 12.3 Captions

```latex
% Caption below the content (typical for figures):
\begin{figure}[htbp]
    \centering
    \includegraphics[width=5cm]{image}
    \caption{This is a figure caption.}
    \label{fig:myfigure}
\end{figure}

% Caption above the content (typical for tables):
\begin{table}[htbp]
    \centering
    \caption{This is a table caption.}
    \label{tab:mytable}
    \begin{tabular}{lc}
        ...
    \end{tabular}
\end{table}
```

Listing 12.2: Captions

### 12.3.1 Caption Styles (caption package)

```latex
\usepackage{caption}

% Change label separator:
\captionsetup{labelsep=period}   % Figure 1. Caption text
\captionsetup{labelsep=quad}     % Figure 1    Caption text
\captionsetup{labelsep=newline}  % Label on separate line
\captionsetup{labelsep=none}     % No label

% Font options:
\captionsetup{font=small}
\captionsetup{font=it}
\captionsetup{labelfont=bf}

% Justification:
\captionsetup{justification=centering}
\captionsetup{justification=raggedright}
\captionsetup{justification=justified}

```

```
% Per-environment:
\captionsetup[figure]{font=small, labelfont=bf}
\captionsetup[table]{font=small, skip=5pt}
```

Listing 12.3: Caption customization

## 12.4 Side Captions

```
\usepackage{sidecap}

\begin{SCfigure}
    \includegraphics[width=5cm]{image}
    \caption{Caption appears beside the figure.}
\end{SCfigure}
```

Listing 12.4: Side captions

## 12.5 Float Barriers

Sometimes floats drift too far from their reference. Use barriers to control this:

```
\usepackage{placeins}

% Force all pending floats to be placed before this point:
\FloatBarrier

% Use in a section command:
\makeatletter
\AtBeginSection{\FloatBarrier}
\makeatother
```

Listing 12.5: Float barriers

## 12.6 Controlling Float Counters

```
% Maximum floats per page:
\renewcommand{\topfraction}{0.9}      % Max fraction of page for top floats
\renewcommand{\bottomfraction}{0.9}   % Max fraction for bottom floats
\renewcommand{\textfraction}{0.05}    % Min fraction for text
\renewcommand{\floatpagefraction}{0.8} % Min fraction of floats for float page

% Maximum number of floats:
\setcounter{topnumber}{4}             % Max top floats per page
\setcounter{bottomnumber}{4}          % Max bottom floats per page
\setcounter{totalnumber}{10}          % Max total floats per page
```

Listing 12.6: Float counters

# Chapter 13

# Essential Packages & Their Uses

## 13.1 Package Management

```latex
% Basic syntax:
\usepackage{packagename}

% With options:
\usepackage[option1, option2]{packagename}

% Multiple packages at once:
\usepackage{package1, package2, package3}
```

Listing 13.1: Loading packages

*Note: Packages must be loaded in the preamble (before* `\begin{document}`*). Load* `hyperref` *last (or near-last) as a general rule.*

## 13.2 Essential Package Reference

| Package | Category | Purpose |
|---|---|---|
| `inputenc` | Encoding | UTF-8 input encoding |
| `fontenc` | Encoding | Font encoding (T1) |
| `geometry` | Layout | Page margins and layout |
| `graphicx` | Graphics | Include images |
| `amsmath` | Math | Advanced math typesetting |
| `amssymb` | Math | Additional math symbols |
| `amsthm` | Math | Theorem environments |
| `booktabs` | Tables | Professional table rules |
| `array` | Tables | Extended column types |
| `tabularx` | Tables | Flexible-width tables |
| `longtable` | Tables | Multi-page tables |
| `multirow` | Tables | Multi-row cells |
| `hyperref` | Links | Hyperlinks and cross-refs |
| `xcolor` | Color | Color support |

| Package | Category | Purpose |
|---|---|---|
| `listings` | Code | Source code formatting |
| `tikz` | Graphics | Programmable drawings |
| `pgfplots` | Graphics | Plots and charts |
| `fancyhdr` | Layout | Headers and footers |
| `caption` | Floats | Caption customization |
| `subcaption` | Floats | Subfigures/subtables |
| `enumitem` | Lists | Custom list formatting |
| `parskip` | Layout | Paragraph spacing |
| `float` | Floats | Improved float placement |
| `siunitx` | Science | SI units and numbers |
| `url` | Links | URL line breaking |
| `fancyvrb` | Verbatim | Enhanced verbatim |
| `multicol` | Layout | Multiple columns |
| `wrapfig` | Floats | Text-wrapped figures |
| `natbib` | Bibliography | Flexible citations |
| `biblatex` | Bibliography | Modern bibliography |
| `glossaries` | Reference | Glossary / acronyms |
| `index` | Reference | Index generation |
| `nomencl` | Reference | Nomenclature lists |
| `appendix` | Structure | Appendix formatting |
| `titlesec` | Structure | Section title formatting |
| `tocloft` | TOC | TOC customization |
| `setspace` | Layout | Line spacing control |
| `ulem` | Text | Underline / strikethrough |
| `textcomp` | Symbols | Additional text symbols |
| `placeins` | Floats | Float barriers |

## 13.3 Package Conflicts and Loading Order

```latex
% 1. Encoding
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}

% 2. Language (babel or polyglossia)
\usepackage[english]{babel}

% 3. Fonts
\usepackage{lmodern}   % Latin Modern fonts

% 4. Math
\usepackage{amsmath, amssymb, amsthm}

% 5. Graphics and drawing
\usepackage{graphicx}
\usepackage{tikz}

% 6. Tables
```

```latex
\usepackage{booktabs, array, tabularx}

% 7. Layout
\usepackage{geometry}
\usepackage{fancyhdr}

% 8. Lists and text
\usepackage{enumitem}
\usepackage{xcolor}

% 9. Code
\usepackage{listings}

% 10. Bibliography
\usepackage{natbib}  % or biblatex

% 11. Hyperref (ALWAYS LAST or near-last)
\usepackage{hyperref}

% 12. Appendix (after hyperref)
\usepackage{appendix}
```

Listing 13.2: Recommended loading order

## 13.4 SI Units (siunitx)

```latex
\usepackage{siunitx}

% Number formatting:
\num{12345.67}                % 12 345.67
\num{1.23e4}                  % 1.23 x 10^4
\num[round-precision=2]{3.14159}  % 3.14

% Units:
\si{kg}                       % kg
\si{m/s}                      % m/s
\si{kg.m.s^{-2}}              % kg m s^-2
\si{\meter\per\second}        % m/s (symbolic)

% Value + unit:
\SI{9.8}{m/s^2}               % 9.8 m/s^2
\SI{100}{\degreeCelsius}      % 100 degrees C
\SI{3.14}{\micro\meter}       % 3.14 micrometer
```

Listing 13.3: siunitx package

Examples: The speed of light is $3 \times 10^8$ m/s. Water boils at 100 °C. The wavelength is 550 nm.

## 13.5 Creating Your Own Package

```latex
% Save as mystyle.sty:
\NeedsTeXFormat{LaTeX2e}
\ProvidesPackage{mystyle}[2024/01/01 My Custom Style]

% Package options:
\DeclareOption{draft}{\typeout{Draft mode}}
\ProcessOptions\relax

% Load required packages:
\RequirePackage{amsmath}
\RequirePackage{graphicx}

% Define commands:
\newcommand{\emphasis}[1]{\textbf{\textit{#1}}}

\endinput
% Then in your document:
% \usepackage{mystyle}
```

Listing 13.4: Creating a package

# Chapter 14

# Custom Commands & Environments

## 14.1 New Commands (\newcommand)

```latex
% Basic: no arguments
\newcommand{\myname}{John Doe}
Hello, I am \myname.          % Output: Hello, I am John Doe.

% With arguments (1 required argument):
\newcommand{\greet}[1]{Hello, #1!}
\greet{Alice}                 % Output: Hello, Alice!

% With multiple arguments:
\newcommand{\add}[2]{#1 + #2}
$\add{a}{b}$                  % Output: a + b

% With optional arguments:
\newcommand{\hello}[2][World]{Hello, #2 from #1!}
\hello{Alice}                 % Output: Hello, Alice from World!
\hello[Bob]{Alice}            % Output: Hello, Alice from Bob!

% Maximum 9 arguments (#1 through #9)
```

Listing 14.1: newcommand syntax

### 14.1.1 Practical Examples

```latex
% Shorthand for common formatting:
\newcommand{\pkg}[1]{\texttt{#1}}
\newcommand{\cmd}[1]{\texttt{\textbackslash#1}}

% Math shortcuts:
\newcommand{\R}{\mathbb{R}}               % Real numbers
\newcommand{\norm}[1]{\left\|#1\right\|} % Norm notation
\newcommand{\abs}[1]{\left|#1\right|}     % Absolute value
\newcommand{\inner}[2]{\langle #1, #2 \rangle} % Inner product
\newcommand{\dd}{\,\mathrm{d}}            % Differential d

```

```
% Usage:
% $f\colon \R^n \to \R$
% $\norm{x} = \sqrt{\inner{x}{x}}$
% $\int_0^1 f(x) \dd x$
```

Listing 14.2: Practical custom commands

Results:

$$f\colon \mathbb{R}^n \to \mathbb{R}, \quad \|x\| = \sqrt{\langle x, x\rangle}$$

## 14.2 Renaming Commands (\renewcommand)

```
% Override existing commands:
\renewcommand{\contentsname}{Contents}          % Rename TOC title
\renewcommand{\figurename}{Fig.}                % Rename figure
    label
\renewcommand{\tablename}{Table}                % Rename table
    label
\renewcommand{\bibname}{References}             % Rename
    bibliography

% Check if command exists:
\providecommand{\mycommand}{default value}    % Define only if not
     already
```

Listing 14.3: renewcommand

## 14.3 New Environments

You can define entire environments (begin/end pairs) with `\newenvironment`:

The syntax is: `\newenvironment{name}[args]{begin-code}{end-code}`

**Example 1** – a simple warning box:

```
\newenvironment{warning}
    {\begin{center}\bfseries Warning: }
    {\end{center}}
```

**Example 2** – a box with a title argument:

```
\newenvironment{infobox}[1]
    {\begin{center}\fbox{\parbox{0.9\textwidth}%
     {\textbf{##1}\par\smallskip}}}
    {\end{center}}
```

## 14.4 Lengths and Counters

### 14.4.1 Defining Lengths

Custom lengths

```
% Create a new length:
\newlength{\mywidth}
\setlength{\mywidth}{5cm}

% Use it:
\hspace{\mywidth}

% Arithmetic on lengths:
\addtolength{\mywidth}{2cm}           % Add 2cm
\settowidth{\mywidth}{Hello World}    % Set to width of text
\settoheight{\myheight}{Text}         % Set to height
\settodepth{\mydepth}{Text}           % Set to depth
```

### 14.4.2 Standard Lengths

| Length | Controls |
|---|---|
| \textwidth | Width of text area |
| \linewidth | Width of current line |
| \columnwidth | Width of column |
| \paperwidth | Width of paper |
| \paperheight | Height of paper |
| \parindent | Paragraph indentation |
| \parskip | Space between paragraphs |
| \baselineskip | Distance between baselines |
| \columnsep | Space between columns |

Table 14.1: Standard LaTeX lengths

### 14.4.3 Counters

```
% Define a new counter:
\newcounter{mycounter}

% Set, step, and use:
\setcounter{mycounter}{0}
\stepcounter{mycounter}       % Increment by 1
\addtocounter{mycounter}{5}   % Add 5

% Display:
\arabic{mycounter}    % 1, 2, 3
\roman{mycounter}     % i, ii, iii
\Roman{mycounter}     % I, II, III
\alph{mycounter}      % a, b, c
\Alph{mycounter}      % A, B, C
```

```latex
\fnsymbol{mycounter} % *, dagger, double-dagger, etc.
```

Listing 14.4: Counters

## 14.5 Conditional Commands

```latex
% ifthen package:
\usepackage{ifthen}

\newcommand{\checkodd}[1]{%
    \ifthenelse{\isodd{#1}}%
        {#1 is odd}%
        {#1 is even}%
}

% Boolean flags:
\newboolean{draftmode}
\setboolean{draftmode}{true}

\ifthenelse{\boolean{draftmode}}%
    {\watermark{DRAFT}}%
    {}
```

Listing 14.5: Conditionals

## 14.6 Boxes

Boxes

```latex
% mbox: prevents line breaks
\mbox{This text stays on one line}

% fbox: framed box
\fbox{Framed text}

% makebox: custom box with alignment
\makebox[5cm][c]{Centered in 5cm}
% Alignment: l=left, c=center, r=right, s=stretch

% parbox: paragraph in a box
\parbox{3cm}{This text wraps inside a 3cm-wide box.}

% minipage: like parbox but more powerful
\begin{minipage}[t]{0.45\textwidth}
    Left column content.
\end{minipage}
\hfill
\begin{minipage}[t]{0.45\textwidth}
    Right column content.
\end{minipage}
```

```
% raisebox: vertically offset text
\raisebox{5pt}{Raised text}
\raisebox{-5pt}{Lowered text}
```

Side-by-side minipage example:

**Left Column**
This content sits in the left minipage. It wraps automatically within the specified width.

**Right Column**
This content sits in the right minipage. Using minipages we can create multi-column layouts anywhere.

# Chapter 15

# Advanced Topics

## 15.1 Tables of Contents for Non-Standard Sections

```latex
% Add unnumbered chapter to TOC:
\chapter*{Preface}
\addcontentsline{toc}{chapter}{Preface}

% Custom TOC depth:
\setcounter{tocdepth}{2}      % Show sections and subsections only
\setcounter{secnumdepth}{3}   % Number up to subsubsection

% Mini table of contents per chapter (minitoc package):
\usepackage{minitoc}
\dominitoc                        % In preamble, after \tableofcontents
\chapter{My Chapter}
\minitoc                          % Shows TOC for this chapter only
```

Listing 15.1: Custom TOC entries

## 15.2 Index

```latex
\usepackage{makeidx}
\makeindex                         % In preamble

% Mark entries:
\index{keyword}
\index{main topic!subtopic}       % Nested index entry
\index{term|textbf}               % Bold page reference

% Special characters:
\index{keyword@\texttt{keyword}}  % Custom sort vs display

% Print the index:
\printindex                        % At end of document

```

```
% Compilation:
% pdflatex main.tex
% makeindex main
% pdflatex main.tex
```

Listing 15.2: Index

## 15.3 Glossary / Acronyms

```
\usepackage[acronym, toc]{glossaries}
\makeglossaries

% Define terms:
\newglossaryentry{latex}{
    name=LaTeX,
    description={A document preparation system}
}

\newacronym{api}{API}{Application Programming Interface}

% Use in text:
\gls{latex}          % First: "LaTeX (A document preparation system)"
                      % Later: "LaTeX"
\gls{api}            % First: "Application Programming Interface (API)"
                      % Later: "API"
\Gls{latex}          % Capitalized first letter

% Print glossary:
\printglossary[type=main]           % Glossary
\printglossary[type=\acronymtype]   % Acronyms
```

Listing 15.3: Glossary

## 15.4 Watermarks

```
\usepackage[firstpage]{draftwatermark}
\SetWatermarkText{CONFIDENTIAL}
\SetWatermarkScale{3}
\SetWatermarkColor[gray]{0.9}
\SetWatermarkAngle{45}

% Or with wallpaper package for images:
\usepackage{wallpaper}
\ThisCenterWallPaper{0.5}{logo.png}
```

Listing 15.4: Watermark

## 15.5 Conditional Compilation

```latex
% Define a flag:
\newif\ifdraft
\drafttrue          % Set to draft mode
% \draftfalse       % Set to final mode

% Use conditionally:
\ifdraft
    \usepackage[draft]{graphicx}  % Show image boxes only
\else
    \usepackage{graphicx}         % Full images
\fi

% Include content conditionally:
\ifdraft
    \include{chapters/draft-notes}
\fi

% Version control:
\newcommand{\version}{1.0}
\newcommand{\releasedate}{2024-01-01}
```

Listing 15.5: Conditional compilation

## 15.6 External Files and Shell Escape

```latex
% Read from external file:
\input{data/variables.tex}        % Insert file content

% Write to file (during compilation):
\newwrite\myoutput
\immediate\openout\myoutput=output.txt
\immediate\write\myoutput{Hello from LaTeX}
\immediate\closeout\myoutput

% Shell escape (compile with --shell-escape):
% \write18{date > date.txt}
% \input{date.txt}
```

Listing 15.6: File operations

## 15.7 Compilation Tools

### 15.7.1 latexmk

```bash
# latexmk automates multi-pass compilation:
latexmk -pdf main.tex              % Compile to PDF
latexmk -xelatex main.tex          % Use XeLaTeX
latexmk -lualatex main.tex         % Use LuaLaTeX
```

```
latexmk -c                          % Clean auxiliary files
latexmk -C                          % Clean all (including PDF)
latexmk -pvc main.tex               % Preview continuously (live
   reload)
```

Listing 15.7: latexmk automation

### 15.7.2 Compilation Engines

| Engine | Command | Best For |
|---|---|---|
| pdfLaTeX | `pdflatex` | Standard documents, English |
| XeLaTeX | `xelatex` | Unicode, system fonts |
| LuaLaTeX | `lualatex` | Advanced scripting, Lua integration |

Table 15.1: LaTeX compilation engines

### 15.7.3 Using System Fonts (XeLaTeX/LuaLaTeX)

```
% Requires XeLaTeX or LuaLaTeX
\usepackage{fontspec}

% Set main font:
\setmainfont{Times New Roman}
\setsansfont{Arial}
\setmonofont{Consolas}

% With options:
\setmainfont[
    BoldFont=Times New Roman Bold,
    ItalicFont=Times New Roman Italic,
    Scale=1.0
]{Times New Roman}
```

Listing 15.8: System fonts with fontspec

## 15.8 Beamer Presentations (Overview)

```
\documentclass{beamer}
\usetheme{Madrid}
\usecolortheme{default}

\title{My Presentation}
\author{Author Name}
\date{\today}

\begin{document}

\begin{frame}
```

```latex
    \titlepage
\end{frame}

\begin{frame}{Outline}
    \tableofcontents
\end{frame}

\begin{frame}{My First Slide}
    \begin{itemize}
        \item First point
        \item Second point
    \end{itemize}
\end{frame}

% Columns:
\begin{frame}{Two Columns}
    \begin{columns}
        \column{0.5\textwidth}
            Left content
        \column{0.5\textwidth}
            Right content
    \end{columns}
\end{frame}

% Overlays (step-by-step reveal):
\begin{frame}{Overlays}
    \begin{itemize}
        \item<1-> First (appears on slide 1+)
        \item<2-> Second (appears on slide 2+)
        \item<3-> Third (appears on slide 3+)
    \end{itemize}
\end{frame}
\end{document}
```

Listing 15.9: Beamer basics

## 15.9 Multilingual Support

```latex
% With babel:
\usepackage[english, french, german]{babel}
% Last language is default. Switch:
\selectlanguage{french}
\selectlanguage{english}

% With polyglossia (XeLaTeX/LuaLaTeX):
\usepackage{polyglossia}
\setdefaultlanguage{english}
\setotherlanguage{french}
\setotherlanguage{german}

\begin{german}
```

```
    Hallo Welt!
\end{german}
```

Listing 15.10: Multilingual documents

## 15.10 Error Handling

| Error | Common Cause |
|---|---|
| `Undefined control sequence` | Misspelled command or missing package |
| `Missing $ inserted` | Used math symbol outside math mode |
| `Extra alignment tab` | Too many `&` in table row |
| `Missing } inserted` | Unmatched braces |
| `Float too large` | Figure/table wider than text area |
| `Overfull \hbox` | Text extends past margin |
| `Underfull \hbox` | Too much whitespace in line |

Table 15.2: Common LaTeX errors

*Note: Always read the `.log` file for detailed error information. The first error reported is usually the root cause.*

# Appendix A

# Quick Reference Card

## A.1 Document Skeleton

```latex
\documentclass[12pt, a4paper]{report}

% === PREAMBLE ===
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{geometry}
\usepackage{graphicx}
\usepackage{amsmath, amssymb}
\usepackage{booktabs}
\usepackage{hyperref}
\usepackage{xcolor}
\usepackage{listings}
\usepackage{tikz}

\title{My Document Title}
\author{Author Name}
\date{\today}

% === BODY ===
\begin{document}
\maketitle
\tableofcontents

\chapter{Introduction}
\label{ch:intro}

\section{Background}
\label{sec:background}

Text with \textbf{bold}, \textit{italic}, and \texttt{monospace}.

Inline math: $E = mc^2$

Display math:
```

```latex
\[ \int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \]

\begin{figure}[htbp]
    \centering
    \includegraphics[width=0.6\textwidth]{myimage}
    \caption{My figure.}
    \label{fig:myfigure}
\end{figure}

\begin{table}[htbp]
    \centering
    \caption{My table.}
    \label{tab:mytable}
    \begin{tabular}{lcc}
        \toprule
        Name & Value & Note \\
        \midrule
        A    & 1     & x \\
        B    & 2     & y \\
        \bottomrule
    \end{tabular}
\end{table}

\bibliographystyle{plain}
\bibliography{references}

\end{document}
```

Listing A.1: Complete document skeleton

## A.2 Symbol Quick Reference

| Symbol | Code | Symbol | Code |
|---|---|---|---|
| $\alpha$ | `\alpha` | $\beta$ | `\beta` |
| $\gamma$ | `\gamma` | $\delta$ | `\delta` |
| $\epsilon$ | `\epsilon` | $\theta$ | `\theta` |
| $\lambda$ | `\lambda` | $\mu$ | `\mu` |
| $\pi$ | `\pi` | $\sigma$ | `\sigma` |
| $\phi$ | `\phi` | $\omega$ | `\omega` |
| $\sum$ | `\sum` | $\prod$ | `\prod` |
| $\int$ | `\int` | $\sqrt{x}$ | `\sqrt{x}` |
| $\frac{a}{b}$ | `\frac{a}{b}` | $\infty$ | `\infty` |
| $\forall$ | `\forall` | $\exists$ | `\exists` |
| $\in$ | `\in` | $\notin$ | `\notin` |
| $\subset$ | `\subset` | $\cup$ | `\cup` |
| $\cap$ | `\cap` | $\emptyset$ | `\emptyset` |
| $\to$ | `\to` | $\Rightarrow$ | `\Rightarrow` |
| $\leq$ | `\leq` | $\geq$ | `\geq` |
| $\neq$ | `\neq` | $\approx$ | `\approx` |
| $\pm$ | `\pm` | $\times$ | `\times` |
| $\cdot$ | `\cdot` | $\ldots$ | `\ldots` |
| $\cdots$ | `\cdots` | $\vdots$ | `\vdots` |

Table A.1: Common math symbols quick reference

## A.3 Useful Commands Cheat Sheet

| Command | Purpose |
|---|---|
| `\documentclass{article}` | Set document type |
| `\usepackage{pkg}` | Load a package |
| `\begin{env}...\end{env}` | Use an environment |
| `\textbf{text}` | Bold text |
| `\textit{text}` | Italic text |
| `\texttt{text}` | Monospace text |
| `\underline{text}` | Underlined text |
| `\textcolor{red}{text}` | Colored text |
| `\footnote{text}` | Footnote |
| `\cite{key}` | Citation |
| `\ref{label}` | Cross reference |
| `\pageref{label}` | Page reference |
| `\label{key}` | Define label |
| `\includegraphics{file}` | Include image |
| `\tableofcontents` | Table of contents |
| `\listoffigures` | List of figures |
| `\listoftables` | List of tables |
| `\appendix` | Start appendix |
| `\today` | Current date |
| `\LaTeX` | LaTeX logo |
| `\TeX` | TeX logo |

Table A.2: Essential commands cheat sheet

## A.4 Compilation Workflow

| Task | Commands |
|---|---|
| Basic compile | `pdflatex main.tex` |
| With BibTeX | `pdflatex main` → `bibtex main` → `pdflatex main` ×2 |
| With glossary | Add `makeglossaries main` after first compile |
| Automated | `latexmk -pdf main.tex` |
| Clean aux | `latexmk -c` |

Table A.3: Compilation workflows